Fairness Preservation under Computational Budgets in Probabilistic Decision Systems

ICT 5143: AFGML Capstone | Problem 9

[*Student 1, roll number*] [*Student 2, roll number*]

[*Evaluation date*]

SMOKE RUN ON SYNTHETIC DATA — NOT EVIDENCE

Research question and decision context

- Context: bank outreach scored by a Bayesian network under a compute limit; affected: customers, outreach teams, serving engineers, auditors.
- $Y = 1$: recorded term-deposit subscription; $D = 1$: selected for *hypothetical* pre-contact outreach (capacity 0.2).
- Question: at a fixed operation budget, does fairness-aware fallback selection give smaller Δ_{EOpp} than a global fallback, utility-only selection and post-budget threshold adjustment?
- Hypothesis: yes, at ≤ 0.01 AP loss; refuted if a Holm-adjusted paired 95% interval contains 0 or is positive.

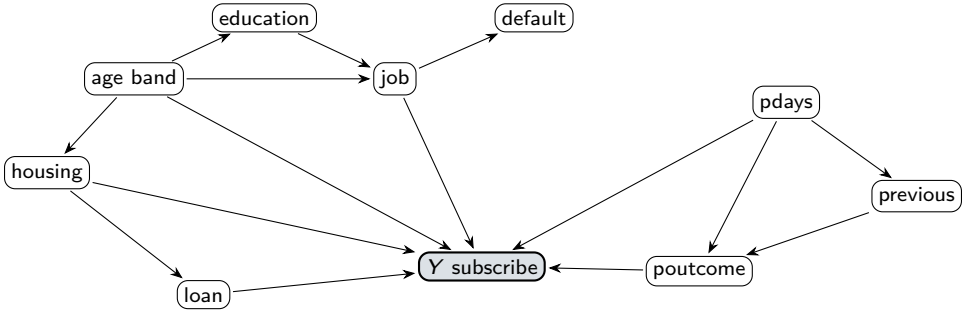
Closest work and proposed contribution

- Established: compute reductions can create group disparities (Das et al., ICML 2024; Hooker et al. 2020) — neural models, accuracy parity.
- Open issue: no verified study found for budget-triggered *approximate PGM inference* and decision fairness (search to be extended).
- Contribution: operation-count budgets with evidence-dependent trips; simulation isolating group-dependent evidence availability (with control); fairness-aware fallback selection.
- Evidence: comparison with global fallback, utility-only selection, Hardt-style post-processing; ablations.

Data and evaluation population

- UCI Bank Marketing (dataset 222), `bank-additional-full.csv`; $Y = 1$ iff `y = yes`; pre-contact variables only (**no `duration`**).
- Protected: age band (< 30, 30–59, ≥ 60) $\times$ marital status (audit only).
- Split: positional 60/20/20 (date-ordered release); test size 1600; no customer ids.
- Model inputs: age band, job, education, default, housing, loan, pdays bin, previous bin, poutcome; `unknown` $\Rightarrow$ unobserved.
- Groups with < 30 relevant observations: exploratory (see backup).

Graphical model and inference



- No latent variables; unobserved evidence plays the role of the mask R. Query $P(Y{=}1 \mid e)$ by VE (min-fill/min-weight/min-degree) and clique tree; cost = factor-entry multiplications + additions.
- Validation: VE vs clique tree vs enumeration: passed, max error 2.2e-16 (tolerance 10^{-6}).

Fairness criterion and proposed method

- Primary: $\Delta_{\mathrm{EOpp}} = \max_g t_g - \min_g t_g$, $t_g = P(D{=}1 \mid Y{=}1, G{=}g)$; a missed responder is the harm. Diagnostics: Δ_{DP}, Δ_{EOdds}, fallback-rate gap, decision flips vs exact, calibration.
- Utility: AP; allowed loss 0.01 (validation-selected).
- Change: if exact elimination would exceed budget B ops $\Rightarrow$ table lookup. Proposed pick: $f^\star = \arg\min_f \max_g \mathrm{flip}_g(f)$ s.t. $\mathrm{AP\ loss}(f) \leq 0.01$.
- Protected attribute at prediction time: age band is a model variable (yes); routing is group-unaware; marital status unused by models.

Experimental protocol

- Baselines: exact; global fallback; threshold adjustment after budgeted inference; fair-adjusted exact; LR; GBT.
- Same variables, splits, tuning; thresholds/tables/selection from train+validation; test frozen.
- Budgets: quantiles of natural validation op counts (primary 75%: 11 ops); wall-clock separate.
- Runs: 5 seeds; paired block-bootstrap (1000 reps, date blocks); Holm over 3 comparisons.
- Mechanism: synthetic networks (width $\times$ availability, with control); ablations; robustness.

Main result and uncertainty

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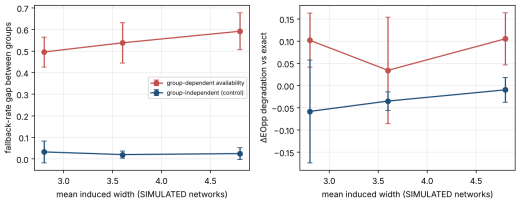
Method	AP ↑	Δ_{EOpp} ↓	Log loss / Brier ↓
Exact (reference)	0.229	0.309	0.288 / 0.078
Global fallback	0.212	0.258	0.292 / 0.079
Proposed (fair selection)	0.231	0.309	0.288 / 0.078
Utility-only selection	0.231	0.309	0.288 / 0.078
Threshold adj. after budget	0.212	0.171	0.292 / 0.079

Paired Δ_{EOpp} effects (proposed − comparator), 95% interval: global +0.000 [-0.125, +0.000] (Holm p=1.000); utility-only +0.000 [+0.000, +0.000] (Holm p=1.000); threshold adj. +0.000 [+0.000, +0.048] (Holm p=1.000).

Answer: [*bounded finding after real run*]

Ablation, robustness and failure case

- Ablations: utility-only selection; global fallback; no evidence-based simplification; generous budget.
- Simulation: fallback-gap difference (dependent vs. independent availability) $+0.516$ vs. control gap 0.026.
- Robustness: elimination orders, densities, discretisation, capacities, masks.
- Failure case / interpretation: [*from real results; sampling vs. scenario uncertainty*]



Conclusion and reproducibility

- Conclusion: [*one supported answer*]
- Limits and ethics: synthetic masks/networks are simulations; rows $\neq$ customers; contacted-only sample.
- Reproduce: `python scripts/run_experiments.py -mode real`, `evaluate.py`, `make_results.py`; configs in `configs/`.
- Student 1: [*contribution and verification*] Student 2: [*contribution and verification*]

Backup: mandatory evidence and references

Requirement-to-result matrix: `docs/requirements_matrix.md`; manifest: `results/manifest.csv`.
Das, Romanelli, Fioretto (ICML 2024); Hooker et al. (2020); Hardt, Price, Srebro (NeurIPS 2016); Moro, Rita, Cortez (UCI 222); Koller & Friedman (2009). Full table: `docs/related_work.md`.

Backup: inference and statistical validation

Normalization, scopes, evidence and exact agreement: `tests/test_cgm.py`, `results/metrics/exp1_exact_agreement.json`. Denominators: `group_denominators.csv`, `group_table.csv`. Bootstrap: fixed-model paired moving blocks (length 50); Holm family = 3 confirmatory comparisons.

Backup: demonstration and individual defence

Command: `python scripts/run_experiments.py -mode smoke -quick -out results/demo` (synthetic, ≈1 min); expected: figures/tables/macros in `results/demo`. Saved fallback trace: `results/smoke/`. Be ready to trace one VE op count, justify EOpp, and critique a claim.